United States Patent Application Publication	20250277724
Kind Code	A1
Publication Date	September 04, 2025
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TISSUE LABELING USING ION EXCHANGE RESIN TO CONTROL BINDING AFFINITY SWEEP

Abstract

A tissue labeling device, method of operation, and a tissue labeling kit enables tissue labeling for microscopy imaging with high labeling depth and uniformity. A tissue labeling device circulates a labeling solution through a tissue sample and concurrently circulates the labeling solution through an ion exchange resin. An ion exchange process increases the binding affinity of the labeling probes in the labeling solution over time as the labeling solution penetrates deeper into the tissue sample. A tissue labeling kit may operate on similar principles. The tissue labeling kit includes a container (e.g., a tube) having an ion exchange resin layer and a hydrogel layer over the ion exchange resin layer. The labeling solution and tissue sample is added to the container to enable the binding affinity sweep in which the hydrogel operates to control the sweep rate of the binding affinity.

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Family ID: **96881095**

Appl. No.: **19/064516**

Filed: **February 26, 2025**

Related U.S. Application Data

us-provisional-application US 63559872 20240229

Publication Classification

Int. Cl.: **G01N1/31** (20060101)

U.S. Cl.:

CPC **G01N1/312** (20130101);

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application claims the benefit of U.S. Patent Provisional Application No. 63/559,872 filed on Feb. 29, 2024, which is incorporated by reference herein.

BACKGROUND
[0002] Tissue labeling is a technique used in microscopy applications to enhance visibility of structures in tissue samples for imaging. Technical advances in tissue preparation and imaging techniques have enabled rapid, high-resolution imaging of large, intact biological samples. However, it remains difficult to achieve uniform labeling of specific biomolecules in these intact samples. Ex vivo delivery of antibodies or molecular dyes relies on simple diffusion of labeling probes (molecules) into the tissue. In many cases antibodies remain the gold standard for specific labeling of protein antigens. However, the relatively large size of immunoglobulin G antibodies (IgGs) means their diffusion into the tissue is slow. This slow diffusion coupled with the high affinity of antibodies often results in the depletion of antibodies from the solution as they diffuse into the sample, and therefore a non-uniform labeling profile. In this case, the majority labeling probes are used up by the antigens on the surface of the sample before they are able to diffuse into the depth of the tissue.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
[0003] FIG. **1** is an example embodiment of a tissue labeling device.
[0004] FIG. **2** is an example embodiment of an electronics system for a tissue labeling device.
[0005] FIG. **3** is a flowchart illustrating an example embodiment of a process for operating a tissue labeling device.
[0006] FIG. **4** is an example embodiment of an ion exchange resin attachment for a tissue labeling device.
[0007] FIG. **5** is an example embodiment of tissue labeling kit for performing tissue labeling.
[0008] FIG. **6** is a flowchart illustrating an example embodiment of a process for performing tissue labeling using a tissue labeling kit.
[0009] FIG. **7** is a plot illustrating a rate of change of binding affinity associated with a tissue labeling process for achieving uniform tissue labeling.
[0010] FIG. **8** is a first set of example images comparing labeling uniformity of a mouse brain using different labeling techniques.
[0011] FIG. **9** is a second set of example images comparing labeling uniformity of a mouse brain using different labeling techniques.
[0012] FIG. **10** is a plot illustrating comparing labeling profiles under different labeling techniques relative to an ideal labeling profile.
DETAILED DESCRIPTION
[0013] The Figures (FIGS.) and the following description describe certain embodiments by way of illustration only. One skilled in the art will readily recognize from the following description that alternative embodiments of the structures and methods illustrated herein may be employed without departing from the principles described herein. Reference will now be made to several embodiments, examples of which are illustrated in the accompanying figures. Wherever

practicable, similar or like reference numbers may be used in the figures and may indicate similar or like functionality.

[0014] The disclosed embodiments include a tissue labeling device, method of operation, and a tissue labeling kit for tissue labeling with high labeling depth. In the disclosed embodiments, the binding affinity of labeling probes are reliably swept in a controlled manner to achieve high uniformity of labeling in large, intact tissues. An ion-exchange resin interacts with the labeling solution to gradually introduce ions into the labeling solution that cause the binding affinity of the labeling probes to increase over time. Thus, the binding affinity may initially be low and may increase as the labeling solution penetrates deeper into the tissue sample, thereby preventing labeling probes from concentrating on the tissue surface and enabling uniform labeling. In an example embodiment, the rate of change of the binding affinity may increase over time such that the binding affinity increases slowly at first and increases more rapidly after the labeling solution penetrates deep into the tissue sample.

[0015] In an embodiment, a device circulates a labeling solution through a tissue sample and concurrently circulates the labeling solution through an ion exchange resin column. The binding affinity of the labeling probes in the labeling solution increase over time as the labeling solution interacts with the ion exchange resin. The device may regulate the rate of change in binding affinity, in part by controlling flow rate of the labeling solution through the ion exchange resin column. The device may also employ electrophoresis to increase diffusion of the labeling probes into the tissue sample and may control temperature within the labeling chamber to achieve the desired effect.

[0016] In another embodiment, a tissue labeling kit includes a container (e.g., a tube) having an ion exchange resin layer and a hydrogel layer over the ion exchange resin layer. When ready to prepare a tissue sample for labeling, the tissue sample is deposited in the container together with a labeling solution that includes labeling probes. The labeling solution initially includes ions that inhibit the binding affinity of the labeling probes and therefore allow the labeling solution to penetrate the tissue sample. Over time, an ion exchange occurs between the ion exchange resin and the labeling solution by diffusion of ions through the hydrogel layer, which operates to gradually increase binding affinity of the labeling probes.

[0017] FIG. **1** is an example embodiment of a tissue labeling device **100**. The tissue labeling device includes a labeling chamber **102**, a labeling solution reservoir **104**, an ion exchange resin chamber **106**, a pump **108**, and a set of channels **112-1**, . . . **112-5** (collectively referenced herein as channels **112**) for enabling fluid flow between the labeling chamber **102**, the labeling solution reservoir **104**, the ion exchange resin chamber **106**, and the pump **108**.

[0018] The labeling solution reservoir **104** comprises a refillable chamber for storing a labeling solution **114**. The labeling solution **114** may include a liquid buffer with molecules probes designed to specifically bind to certain target proteins, organelles, or other biomolecules in the tissue sample **110**. Examples of labeling probes may include fluorophore conjugated IgG antibodies, nuclear dyes, fluorophore conjugated lectins, fluorophore conjugated oligos, or lipophilic dyes. Different types of labeling solutions **114** with different labeling probes may be used dependent on the type of tissue sample **110** and desired targets. The labeling solution **114** may initially include a significant concentration of ions (i.e. charged molecules) such as $H+$ or $OH-$ that affect pH of the labeling solution **114** and inhibit binding affinity of the labeling probes. The labeling solution may also include d-sorbitol, amino acids, sodium deoxycholate, charged binding inhibitors, and/or other components. In other examples, the labeling solution **114** may initially include one or more binding affinity inhibitors such as, for example, weak acids or bases, ionic detergents, or salts. Depending on the type of tissue for labeling, the labeling solution **114** may comprise an initially high pH or an initially low pH, either of which may inhibit binding affinity.

[0019] The ion exchange resin chamber **106** comprises a chamber for storing an ion exchange resin **116**. The ion exchange resin **116** includes molecules that interact with the labeling solution **114** to neutralize the pH of the labeling solution **114**, which increases the binding affinity of the labeling probes. For example, the ion exchange resin **116** may operate to remove ions from the labeling solution **114** and/or introduce ions to the labeling solution **114**. For example, in some reactions, a labeling solution **114** with an initially high pH (i.e. basic) may be lowered to neutral or near neutral as it interacts with the ion exchange resin **116**. In other examples, a labeling solution **114** with an initially low pH (i.e., acidic) may be raised to neutral or near neutral as it interacts with the ion exchange resin **116**. Once neutralized, the labeling solution **114** may have a pH of 7.0 at 25° C. or within a range sufficient to achieve the appropriate binding affinity and corresponding labeling uniformity suitable for the imaging application (e.g., a range of 6.9-7.1, 6.8-7.2, or other suitable range). Depending on temperature of the labeling solution **114** during the reaction, a different pH or pH range may correspond to a neutral pH suitable for achieving the binding affinity and corresponding labeling uniformity suitable for the imaging application. The amount and type of resin **116** employed to transform the low binding affinity labeling solution into high binding affinity labeling solution may be determined dependent on various factors such as the volume and type of labeling solution, pump speed, temperature, or other factors.

[0020] The labeling chamber **102** comprises a structure for removably holding a tissue sample **110** and for enabling fluid flow of the labeling solution **114** through the labeling chamber **102**. In an example implementation, the tissue sample **110** may be placed in a sample cup having a nanoporous membrane **128** that can be removably inserted into the labeling chamber **102**. The nanoporous membrane **128** may operate to contain labeling probes in a defined volume or to selectively allow diffusion of labeling solution components into and out of the sample cup.

[0021] In an embodiment, the labeling chamber **102** may include a set of electrodes **118** for generating an electric field within the labeling chamber **102**. The electric field operates to drive charged labeling probes into the tissue sample **110**, which may increase labeling speed and uniformity. Furthermore, in examples where the labeling solution **114** has an initially high pH to inhibit binding affinity, the electrolytic reaction may produce an acidic byproduct that reduces the pH of the labeling solution and may operate in conjunction with the ion exchange resin to increase binding affinity over time. For example, in one type of labeling solution **114**, d-sorbitol (a component of the solution) breaks down at the electrodes into an acid which neutralizes the initially high pH while the detergent diffuses through a nanoporous membrane **128**. This electrolytic reaction may be controlled together with the ion exchange process to achieve a controlled rate of increase in binding affinity.

[0022] The labeling chamber **102** may furthermore include one or more temperature control elements **122** (e.g., temperature sensor, heating element, and/or cooling element) that may operate to control temperature in the labeling chamber **102** within a desired temperature range suitable for labeling. For example, the temperature control elements **122** may operate to compensate for Joule heating caused by the electrophoresis.

[0023] The pump **108** operates to pump the labeling solution **114** between the labeling solution reservoir **104**, the ion exchange resin chamber **106**, the labeling chamber **102**, and the various channels **112**. In one such implementation, the pump **108** operates to circulate the labeling solution through two parallel circulation paths **124**, **126**: a first circulation path **124** between the labeling solution reservoir and the ion exchange resin chamber **106**, and a second circulation path **126** between the labeling solution reservoir **104** and the labeling chamber **102**. For example, in one implementation, the pump **108** includes an inlet **132** coupled to the labeling solution reservoir **104** via a channel **112-3** and a first outlet **130** for pumping the labeling solution **114** into the ion exchange resin chamber **106** via a channel **112-5**. The labeling solution **114** returns to the labeling solution reservoir **104** via a channel **112-4**. The pump **108** also includes a second outlet **134** for pumping the labeling solution into the labeling chamber **102** via a channel **112-2**. In this circulation path **126**, the labeling solution **114** returns to the labeling solution reservoir **104** via a channel **112-1**.

[0024] The tissue labeling device **100** may furthermore include an electronics system **200** that enables various electronic control of the device **100**. The electronics system **200** may be implemented using one or more integrated circuits mounted to a printed circuit board, such as one or more microcontrollers, one or more general purpose processors (CPUs), one or more display processors (GPUs), one or more field-programmable gate arrays (FPGAs), one or more application specific integrated circuits (ASICs), one or more memory chips, and various supporting circuits including analog circuits, digital circuits, power control circuits, etc. One or more aspects of the electronics system **200** may be implemented in software and/or firmware. Here, the electronic system **200** may include a non-storage computer-readable storage medium that stores computer-executable instructions that when executed by one or more processors cause the processors to carry out the functions described herein.

[0025] The tissue labeling device **100** may be operated with various controllable operating parameters that affect the rate of permeability of the labeling probes into the tissue sample **110** and the rate of change in binding affinity of the labeling probes. These various parameters may be controlled to achieve labeling uniformity for tissue samples **110** of various types, thicknesses, and target structures. For example, the rate of change binding affinity may be controlled based on the composition and volume of the ion exchange resin **116**, the composition and volume of the labeling solution **114**, the flow rate of the pump **108**, the voltage and/or current output of the electrodes **118** affecting the electrolytic reaction, the temperature in the labeling chamber **102**, the permeability of the membrane **128** around the tissue sample **110**, environmental factors such as humidity, bubble formation or instability of other liquid components, or other factors. The electronic system **200** may enable electronic control of the pump **108**, the electrodes **118**, and the temperature control element **122** to variably control these elements in conjunction with the choice of labeling solution **114**, ion exchange resin **116**, and other physical components affecting the reaction.

[0026] FIG. **2** is an example embodiment of an electronic system **200** for a tissue labeling device **100**. The electronic system **200** may include one or more input/output devices **202**, a mode controller **208**, a pump controller **204**, an electrophoresis controller **210**, a temperature controller **206**, and a power module **212**.

[0027] The I/O devices **202** may include one or more buttons, dials, knobs, touchscreens, remote controllers, or other mechanisms for providing user input to the tissue labeling device **100**. The I/O devices **202** may control various settings or operations such as starting or stopping the pump, controlling the pump speed, controlling temperature, controlling voltages and/or currents for electrophoresis, controlling time durations for labeling processes, etc. The I/O devices **202** may furthermore include one or more visual or audio indicators such as a light emitting diode (LED), liquid crystal display (LCD) screen, speaker, or other output device that may provide visual or audio information about device status, current settings, etc.

[0028] The mode controller **208** may control an operating mode of the tissue labeling device **100**. For example, the mode controller **208** may control switching between one or more different labeling modes that may be configured with different settings (e.g., pump speed, time duration, etc.) and may be suitable for labeling different types of tissue. In some implementations, the tissue labeling device **100** may also be operable in a tissue clearing mode. In this mode, a clearing buffer may be used in place of the labeling solution **114** to perform tissue clearing prior to labeling.

[0029] The pump controller **204** may control operation of the pump **108**. For example, the pump controller **204** may control and on/off state of the pump, a pump speed, an operating duration, or other setting of the pump **108**. In an embodiment, the pump controller **204** may operate in conjunction with the mode controller **208** to control the pump according to the selected mode.

[0030] The electrophoresis controller **210** may control electrophoresis applied in the labeling chamber **102**. For example, the electrophoresis controller **210** may control an on/off state of the electrodes **118** in the labeling chamber **102**, a voltage and/or current applied in the labeling chamber **102**, a duration of voltage and/or current, or other operating conditions. The electrophoresis controller **210** may operate in conjunction with the mode controller **208** to control the electrodes **118** according to the selected mode.

[0031] The temperature controller **206** may control temperature within the labeling chamber **102**. For example, the temperature controller **206** may receive temperature measurements from one or more temperatures sensors within the labeling chamber **102** and control a heating element (e.g., a thermoelectric heater) and/or a cooling element (e.g., a thermoelectric cooler) to adjust temperature to a desired set temperature or temperature range.

[0032] The power module **212** supplies power to the pump **108** and various components of the electronic system **200**. The power module **212** may convert power from a power source (e.g., an outlet or battery) to appropriate voltages and/or currents associated with the elements of the electronic system **200** and the pump **108**.

[0033] FIG. **3** is a flowchart illustrating operation of the tissue labeling device **100** of FIG. **1**. The tissue labeling device **100** holds **302** the sample tissue in the labeling chamber **102**, stores **304** labeling solution **114** in a labeling solution reservoir **104**, and stores **306** an ion exchange resin **116** in a resin chamber **106**. The labeling solution **114** and ion exchange resin **116** may be replaced as needed (e.g., after each labeling operation). Once loaded, the tissue labeling device **100** controls **308** operation of the pump **108** to circulate the labeling solution **114** through the resin chamber **106** and through the labeling chamber **102**. The tissue labeling device **100** may also control other elements such as voltage of the electrodes **118**, heating or cooling of a temperature control element **122**, or other operating parameters to set conditions suitable for tissue labeling. As described above, this process causes the binding affinity of the labeling solution **114** to increase over time such that the labeling probes may initially penetrate the tissue sample **110** without binding at the surface (due to initially low binding affinity) and then bind to the tissue sample **110** with high uniformity.

[0034] FIG. **4** illustrates an example embodiment of a resin column attachment **410** that may be attached to a tissue labeling device **400**. In this embodiment, the tissue labeling device **400** may comprise similar components to the tissue labeling device **100** of FIG. **1** but can optionally operate without the ion exchange resin chamber **106** and ion exchange resin **116**. In the standalone operating configuration, the labeling solution reservoir **104** and the pump **108** may include a set of plugs **454**, **452** for sealing the labeling solution reservoir **104** and the pump **108** when the resin column attachment **410** is not attached. In this operating configuration, the pump **108** circulates the labeling solution **114** through the labeling chamber **102** as described above. An appropriate labeling solution **114** may be used that has a binding affinity that increases over time in response to the electrophoresis in the labeling chamber **102**. For example, the labeling solution **114** in this configuration may comprise a solution initially having high pH and high concentration of detergents, resulting in an initially low binding affinity. Over time, the electrolytic reaction produces an acidic byproduct, reducing the pH while the detergent diffuses through a semipermeable membrane holding the tissue sample **110** to decrease its concentration and increase the binding affinity to high levels by the end of the labeling run.

[0035] The plugs **454**, **452** may be removable to enable attachment of the resin column attachment **410**, which includes the ion exchange resin **116** and a set of fluid channels **112-4**, **121-5** for connecting to the labeling solution reservoir **104** and the pump **108** respectively, resulting in a configuration similar to FIG. **1**.

[0036] FIG. **5** illustrates an example embodiment of a labeling kit **500** for labeling a tissue sample **110**. The labeling kit **500** may comprise a container **510** (such as laboratory conical tube) that contains an ion exchange resin **502** as a base layer, and a hydrogel layer **504** on top of the resin. The hydrogel layer **504** may comprise, for example, agarose, polyacrylamide, alginate, phytagel, gelatin, or a combination thereof. The container **510** may initially include aqueous solution **506** (e.g., distilled water) over the hydrogel **504** to prevent the hydrogel **504** from drying out. The labeling kit **500** may also include the labeling solution **508** in a separate container **520**.

[0037] To perform labeling, the aqueous solution **506** may be removed from the container **510** and the tissue sample **110** may be placed in the container **510** together with the labeling solution **508**. The ion exchange process will then occur passively in the container **510** with the ion exchange resin **502** exchanging ions with the labeling solution **512** to increase binding affinity of the labeling probes over time. The hydrogel layer **504** acts as a barrier to diffusion, thereby controlling the rate of ion exchange. The specific rate of change may be dependent on the thickness and composition of the hydrogel layer **504** (e.g., concentration or other parameters), the composition and volume of the ion exchange resin **502**, the composition and volume of the labeling solution **508**, the characteristics of the tissue sample **110**, environmental factors such as temperature, or other conditions. The labeling solution **512** may initially have relatively low binding affinity when added to the container **510** (e.g., by virtue of it having high or low pH) and thus the labeling probes can diffuse into the tissue sample **110** with relatively little binding. Over time, the labeling solution **512** diffuses through the hydrogel layer **504** and exchanges ions with the resin **502** to slowly increase the binding affinity of the labeling probes (e.g., by decreasing pH to neutral or near neutral when the labeling solution **512** starts with a relative high pH (basic), or by increasing pH to neutral or near neutral when the labeling solution **512** starts with a relatively low pH (acidic)). Once all the ions are exchanged, the binding affinity of the labeling probes will be high to enable uniform labeling.

[0038] In another embodiment, a tissue labeling kit may include a container **510** with separately packaged ion exchange resin **502**, hydrogel **504**, and labeling solution **508**. In this embodiment, a technician may be responsible for measuring out and adding suitable volumes of the ion exchange resin **502**, hydrogel **504**, and the labeling solution **508** to perform labeling of a tissue sample **110**.

[0039] FIG. **6** is an example embodiment of a process for facilitating tissue labeling using a tissue labeling kit **500**. A technician obtains **602** a container **510** including an ion exchange resin **502** and a hydrogel **504** layered over the ion exchange resin. The container **510** may be obtained from a supplier in this form, or the resin **502** and hydrogel **504** may be measured out and added by the technician. If the container **510** comes prefilled with distilled water over the hydrogel layer **504**, this step may furthermore include removing the distilled water. The technician may then add **604** the labeling solution **508** over the hydrogel **504**, and add **606** the tissue sample **110** to the labeling solution **508**. The technician may furthermore control **608** temperature within a suitable temperature range during the ion exchange and labeling reaction. The technician may then remove the tissue sample **110** (now labeled) for imaging using a light sheet microscope or other microscope device. In further embodiments, the above-described process may be performed in automated manner (e.g., by a robotic system) instead of by a human technician.

[0040] FIG. **7** is an example plot **700** showing an increase in binding affinity **702** over time **704** in an example labeling operation using the above-described techniques. As shown, the binding affinity **702** starts relatively low (or at zero in some embodiments). The labeling probes penetrate deeper into the tissue sample **110** over time while the binding affinity is still relatively low and only initially gradually increasing. Over time, the binding affinity **702** increases with increasing rate of change such that the binding affinity becomes relative high. The specific shape of the curve can be controlled such that the high level of binding affinity occurs after sufficient time **704** where high depth of penetration has occurred, thereby achieving uniform labeling.

[0041] FIG. **8** are example sets of image **810**, **820** showing anti-lamin B**1** antibody labeling of an intact mouse brain sample. The standard labeling image **810** was obtained using an electrophoretic sweeping technique without use of the ion exchange resin. The high uniformity labeling image **820** was obtained using the labeling technique of the present disclosure in which an ion exchange resin interacts with the labeling solution to increase binding affinity. As can be seen, labeling uniformity is significantly improved in the high uniformity labeling **820**, particularly in the central regions of the brain. This relatively higher labeling uniformity allows for more accurate imaging of the sample, particularly in the interior regions.

[0042] FIG. **9** is another example set of images **910**, **920** comparing labeling techniques. In this

example, the images **910**, **920** show anti-parvalbumin antibody labeling of an intact mouse brain sample. The high uniformity labeling image **920** shows relatively higher signal strength and uniformity, particularly in the cerebellum region at the bottom of the image.
[0043] FIG. **10** is a plot illustrating comparing labeling profiles for Lamin B**1** in the cerebellum under the standard labeling technique and the high uniformity labeling technique. The plot represents normalized intensity of pixels in images obtained from an imaging device (such as a light sheet microscope or other microscope device) for different distances from the edge of a tissue sample. The ideal labeling profile **1010** shows a constant normalized intensity for a region within the granule layer of the sample indicating uniform binding of labeling probes within his region. The high uniformity labeling profile **1014** (achieved using the high uniformity labeling technique described herein) achieves labeling uniformity close to the ideal labeling profile **1010** and significantly improved relative to the standard labeling profile **1012**.
[0044] The foregoing description of the embodiments has been presented for the purpose of illustration; it is not intended to be exhaustive or to limit the embodiments to the precise forms disclosed. Persons skilled in the relevant art can appreciate that many modifications and variations are possible in light of the above disclosure.
[0045] The language used in the specification has been principally selected for readability and instructional purposes, and it may not have been selected to delineate or circumscribe the inventive subject matter. It is therefore intended that the scope is not limited by this detailed description, but rather by any claims that issue on an application based hereon. Accordingly, the disclosure of the embodiments is intended to be illustrative, but not limiting, of the scope of the invention.

Claims

1. A tissue labeling device, comprising: a labeling chamber for holding a tissue sample; a labeling solution reservoir for containing a labeling solution; a resin chamber for containing an ion exchange resin; a pump to circulate the labeling solution from the labeling solution reservoir through the resin chamber and to circulate the labeling solution through the labeling chamber; and wherein a binding affinity of labeling probes in the labeling solution increases as the labeling solution circulates through the resin chamber based on an ion exchange with the ion exchange resin.
2. The tissue labeling device of claim 1, wherein the binding affinity of the labeling probes increases over time in concurrence with increasing depth of penetration of the labeling solution into the tissue sample.
3. The tissue labeling device of claim 2, wherein a rate of increase of the binding affinity of the labeling probes increases over time such that the rate of change is relatively slower when the labeling solution has a lower depth of penetration into the tissue sample and the rate of increase is relatively higher when the labeling solution has a higher depth of penetration into the tissue sample.
4. The tissue labeling device of claim 1, wherein the labeling chamber includes an electrode to apply an electrical current through the labeling solution in the labeling chamber.
5. The tissue labeling device of claim 1, wherein the tissue labeling device further comprises: a temperature sensor to sense a temperature of the labeling solution in the labeling chamber; a temperature control element to generate heating or cooling in the labeling chamber; and a controller to control the temperature control element based on the temperature to maintain the labeling solution in the labeling chamber within a target temperature range.
6. The tissue labeling device of claim 1, further comprising: a first fluid channel between the labeling solution reservoir and the pump; a second fluid channel between the pump and the resin chamber; a third fluid channel between the resin chamber and the labeling solution reservoir; a fourth fluid channel between the pump and the labeling chamber; and a fifth fluid channel between the labeling chamber and the labeling solution reservoir.

7. The tissue labeling device of claim 1, further comprising: a controller to control operation of the tissue labeling device in a first operating mode for tissue clearing, and to control operation of the tissue labeling device in a second operation mode for tissue labeling.
8. The tissue labeling device of claim 1, wherein the labeling solution comprises an initially high pH solution.
9. The tissue labeling device of claim 8, wherein the ion exchange resin interacts with the labeling solution to neutralize the initially high pH of the labeling solution.
10. A tissue labeling kit, comprising: a tube containing: an ion exchange resin layer; a hydrogel over the ion exchange resin layer; and an aqueous solution over the hydrogel in the tube; and a labeling solution container containing a labeling solution, wherein the labeling solution includes labeling probes that bind to target molecules of a tissue sample with a binding affinity that increases when the labeling solution interacts with the ion exchange resin in an ion exchange process.
11. The tissue labeling kit of claim 10, wherein the hydrogel comprises at least one of: agarose, polyacrylamide, alginate, phytagel, gelatin, or a combination thereof, and wherein the hydrogel operates to control a rate of the ion exchange process.
12. The tissue labeling kit of claim 10, wherein the labeling solution comprises an initially high pH solution.
13. The tissue labeling kit of claim 12, wherein the ion exchange resin interacts with the labeling solution to neutralize the initially high pH of the labeling solution.
14. The tissue labeling kit of claim 10, wherein a rate of increase of the binding affinity increases over time.
15. A method for performing tissue labeling in a tissue labeling device, the method comprising: holding a tissue sample in a labeling chamber; storing labeling solution in a labeling solution reservoir; storing an ion exchange resin in a resin chamber; and controlling a pump to circulate the labeling solution from the labeling solution reservoir through the resin chamber and to circulate the labeling solution through the labeling chamber, such that a binding affinity of labeling probes in the labeling solution increases as the labeling solution circulates through the resin chamber based on an ion exchange with the ion exchange resin.
16. The method of claim 15, wherein the binding affinity of the labeling probes increases over time in concurrence with increasing depth of penetration of the labeling solution into the tissue sample.
17. The method of claim 15, wherein a rate of increase of the binding affinity of the labeling probes increases over time such that the rate of change is relatively slower when the labeling solution has a lower depth of penetration into the tissue sample and the rate of increase is relatively higher when the labeling solution has a higher depth of penetration into the tissue sample.
18. The method of claim 15, further comprising: controlling a set of electrodes in the labeling chamber to apply an electrical current through the labeling solution in the labeling chamber.
19. The method of claim 15, further comprising: sensing, by a temperature sensor, a temperature of the labeling solution in the labeling chamber; and generating heating or cooling in the labeling chamber based on the temperature to maintain the labeling solution in the labeling chamber within a target temperature range.
20. The method of claim 15, further comprising: operating the tissue labeling device in a tissue clearing mode to clear the tissue sample prior to tissue labeling.